

2. If you have a remote control that can turn on the light and turn off the light. Please design a *LightOnCommand* class and a *LightOffCommand* that implement the *Command* interface based on the command pattern. The *Command* interface contains one method named *executeCommand()*. Write a *RemoteControl* class and a *Light* class. The *RemoteControl* class contains two methods *setCommand()* and *invokeCommand()*. Write a *Client* class to test the result.

Client

```
package experiment17_2;

public class Client {

    public static void main(String[] args) {

        Light light = new Light();

        Command lightOnCommand = new
LightOnCommand(light);

        Command lightOffCommand = new
LightOffCommand(light);

        RemoteControl remoteControl = new
RemoteControl();

        remoteControl.setCommand(lightOnCommand);
        remoteControl.invokeCommand();

        remoteControl.setCommand(lightOffCommand);
        remoteControl.invokeCommand();
    }
}
```

```
}  
}
```

Command

```
package experiment17_2;  
  
interface Command {  
    void executeCommand();  
}
```

Light

```
package experiment17_2;  
  
class Light {  
    public void turnOn() {  
        System.out.println("The light is ON.");  
    }  
  
    public void turnOff() {  
        System.out.println("The light is OFF.");  
    }  
}
```

LightOffCommand

```
package experiment17_2;
```

```
class LightOffCommand implements Command{  
    private Light light;  
    public LightOffCommand(Light light){  
        this.light = light;  
    }  
  
    @Override  
    public void executeCommand() {  
        light.turnOff();  
    }  
}
```

LightOnCommand

```
package experiment17_2;  
  
class LightOnCommand implements Command{  
    private Light light;  
    public LightOnCommand(Light light){  
        this.light = light;  
    }  
  
    @Override  
    public void executeCommand() {
```

```
        light.turnOn();  
    }  
}
```

RemoteControl

```
package experiment17_2;  
  
class RemoteControl {  
    private Command command;  
    public void setCommand(Command command){  
        this.command = command;  
    }  
    public void invokeCommand(){  
        if(command != null){  
            command.executeCommand();  
        }else{  
            System.out.println("No command set.");  
        }  
    }  
}
```

Output

```
D:\JDK\bin\java.exe "-javaagent:D:\idea\IntelliJ\bin\idea_rt.jar" -jar D:\JDK\bin\java.exe
```

```
The light is ON.
```

```
The light is OFF.
```

```
Process finished with exit code 0
```